BCSE429P

Cyber Physical Systems Design Lab

Laboratory 1 - MQTT Using Node JS with HiveMQ Serverless Cloud

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Question:

Design a CoAP protocol using NodeJS that is deployed in an industrial environment to check the Air pollution parameters, namely (CO, CO2, NOx, PM and HC). Identify all of these parameters' safer levels and randomise a range (have a minimum and maximum value). Whenever the operator requests a pollution parameter, it should display the corresponding random values chosen from the range <Min, Max>.

Use the Coap client as Californium browser. Follow the instructions given below and check the video for the source code explanation.

Open one command prompt and run the command node coap.js

Open another command prompt and run the following command

C:/> java --module-path C:\Users\tspra\Downloads\javafx-sdk-17.0.16\lib --add-modules javafx.controls,javafx.fxml -jar cf-browser-3.6.0.jar

A browser will open and go to the Menu->CoAP Details -> Accept -> Application/JSON

And in the URL, input as per the following:

coap://localhost:5683/nox (Nitrous Oxide)

coap://localhost:5683/pm (Particle Matter)

coap://localhost:5683/hc (Hydrocarbons)

coap://localhost:5683/co (Carbon Monoxide)

coap://localhost:5683/co2 (Carbon Dioxide)

Code:

```
var coap = require('coap');

function randomInt(min, max) {
    return Math.floor(Math.random() * (max - min + 1)) + min;
}

var portNumber = 5683;

coap.createServer(function (req, res) {

    console.log("Request Received:", req.url);

    if (req.headers['Accept'] != 'application/json') {
        res.code = '4.06';
        return res.end();
    }

    switch (req.url) {

        case "/co":
            displayOutput(res, {
                Parameter: "Carbon Monoxide (CO)",
                Unit: "ppm",
                SafeRange: "0-9",
                Value: randomInt(0, 9)
            });
            break;

        case "/co2":
            displayOutput(res, {
                Parameter: "Carbon Dioxide (CO2)",
                Unit: "ppm",
                SafeRange: "400-1000",
                Value: randomInt(400, 1000)
            });
            break;
```

```
        case "/nox":
            displayOutput(res, {
                Parameter: "Nitrogen Oxides (NOx)",
                Unit: "ppb",
                SafeRange: "0-53",
                Value: randomInt(0, 53)
            });
            break;

        case "/pm":
            displayOutput(res, {
                Parameter: "Particulate Matter (PM2.5)",
                Unit: "µg/m³",
                SafeRange: "0-35",
                Value: randomInt(0, 35)
            });
            break;

        case "/hc":
            displayOutput(res, {
                Parameter: "Hydrocarbons (HC)",
                Unit: "ppm",
                SafeRange: "0-160",
                Value: randomInt(0, 160)
            });
            break;

        default:
            res.code = '4.04';
            res.end("Invalid Resource");
    }

}).listen(portNumber);

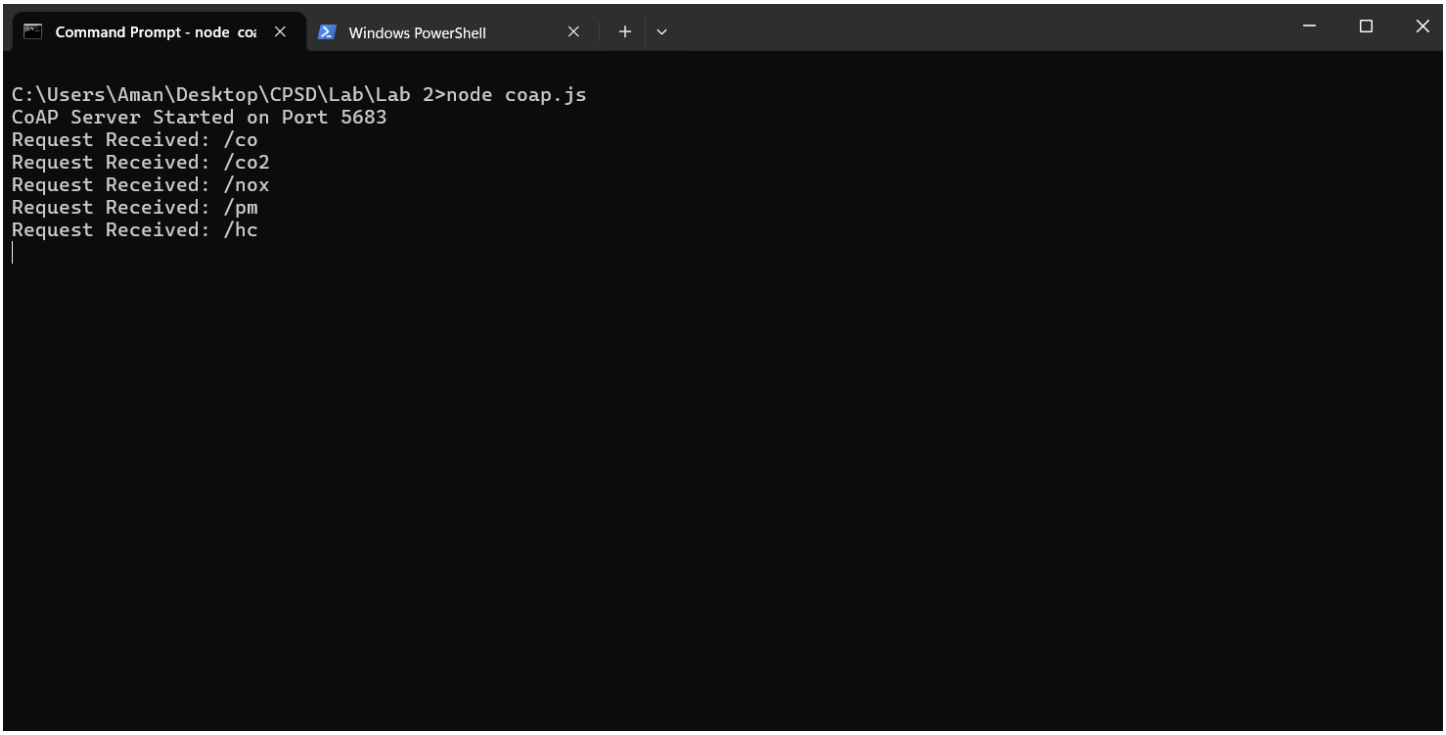
console.log("CoAP Server Started on Port 5683");

function displayOutput(res, content) {

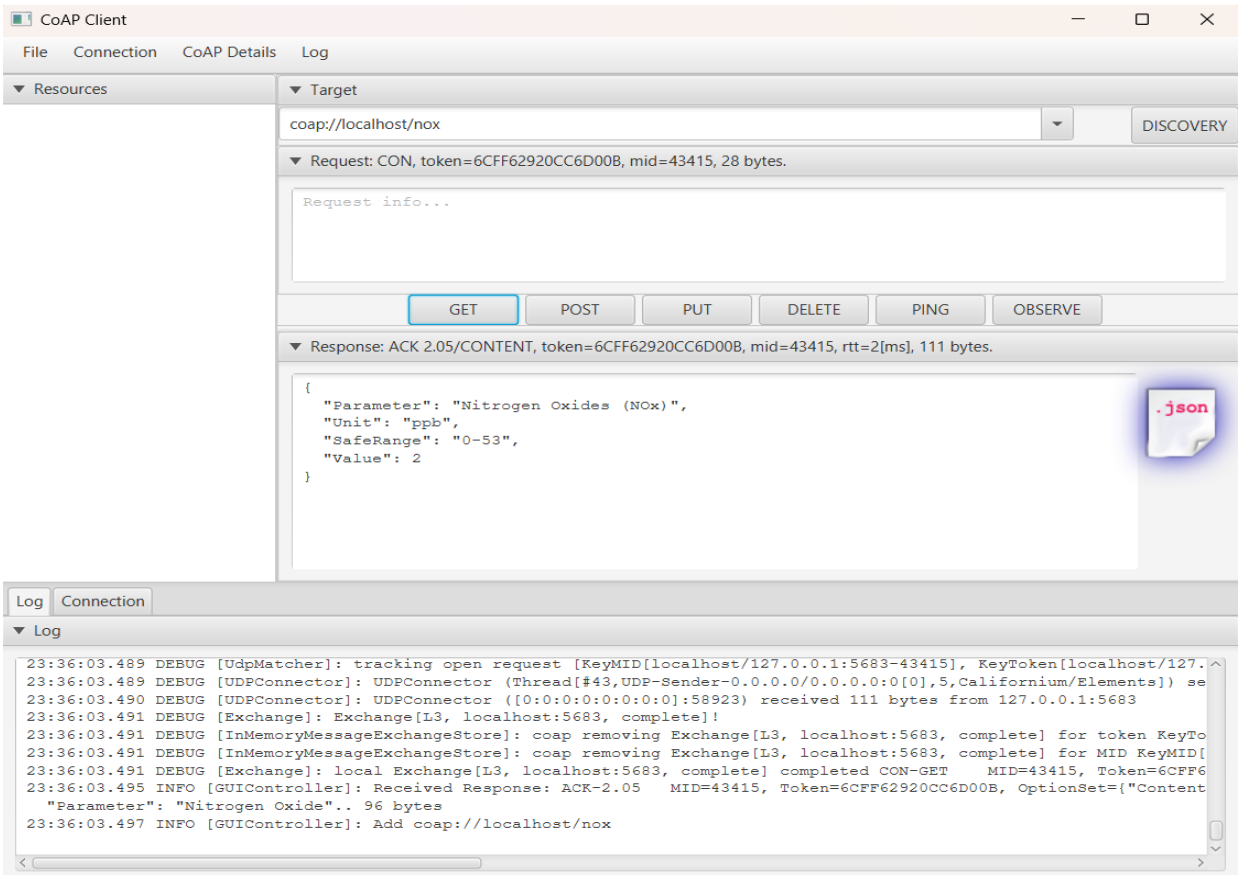
    res.setOption('Content-Format', 'application/json');
    res.code = '2.05';
    res.end(JSON.stringify(content, null, 2));

}
```

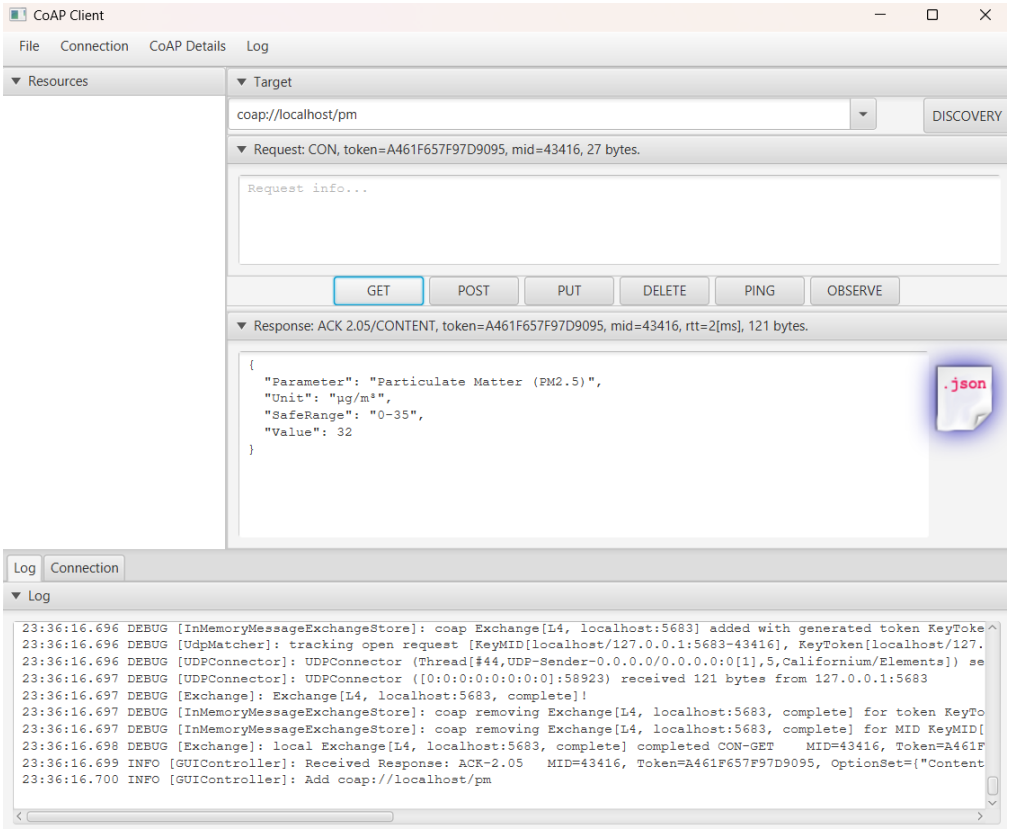
Output:



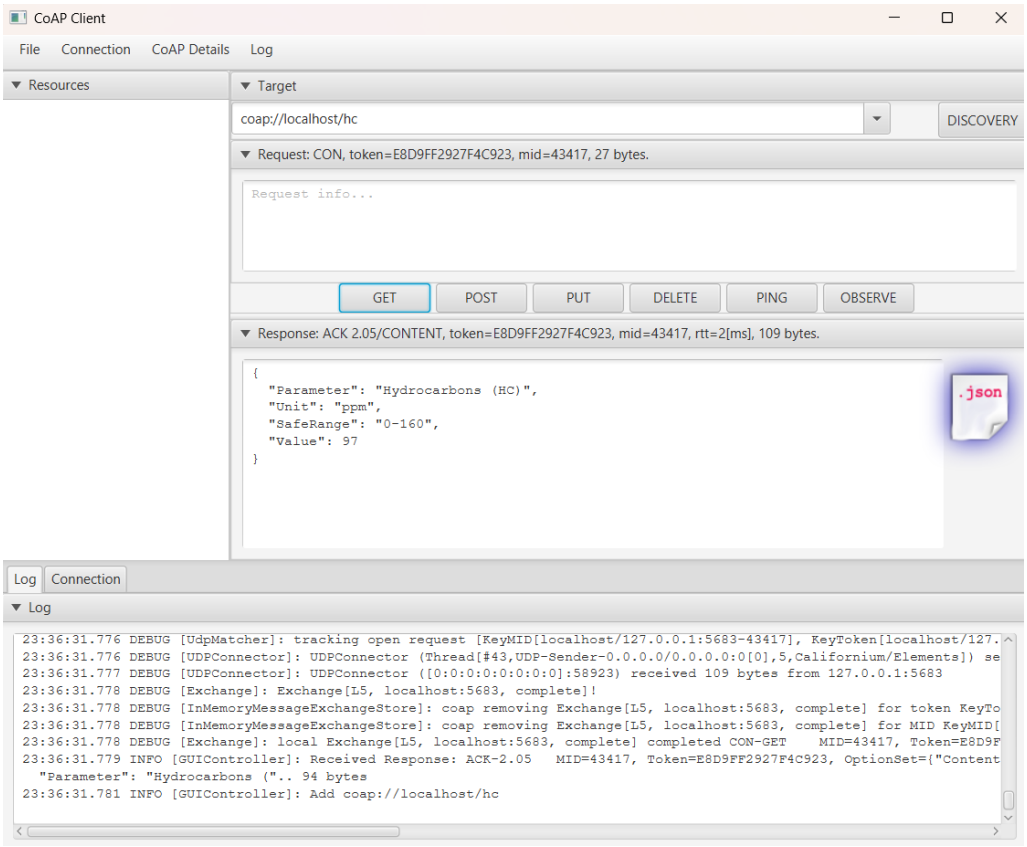
coap://localhost:5683/nox (Nitrous Oxide)



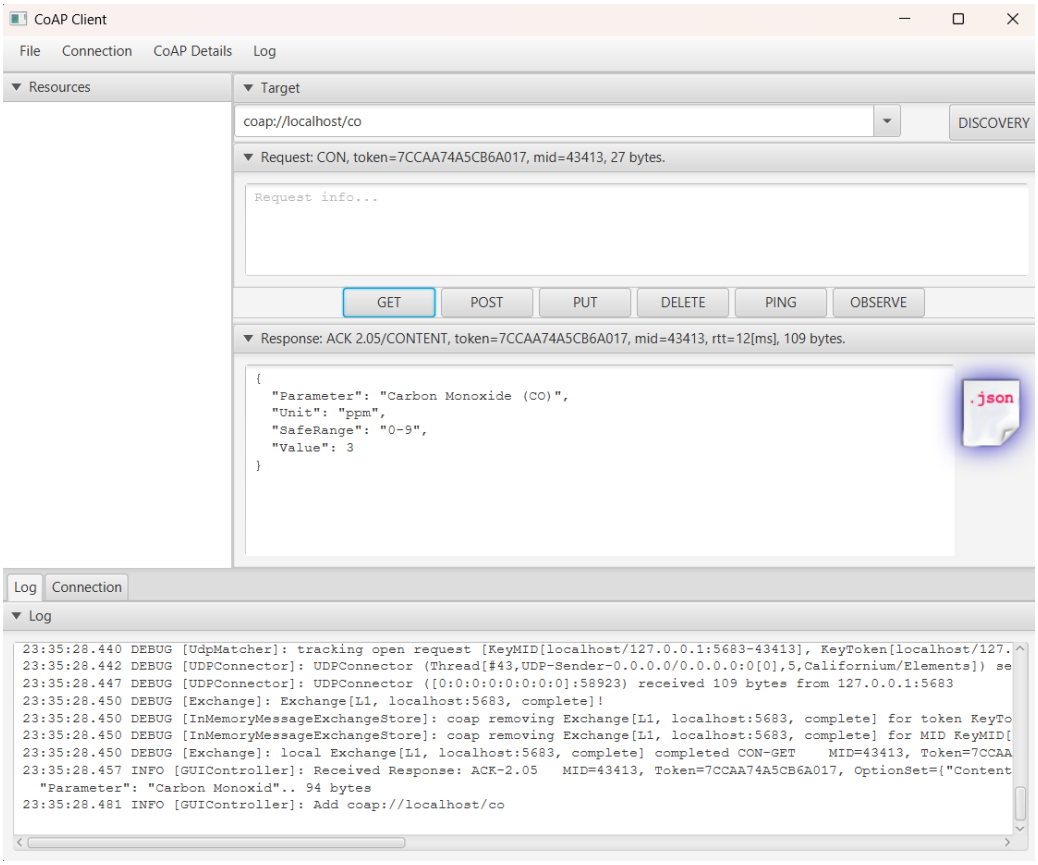
coap://localhost:5683/pm (Particle Matter)



coap://localhost:5683/hc (Hydrocarbons)



coap://localhost:5683/co (Carbon Monoxide)



coap://localhost:5683/co2 (Carbon Dioxide)

